

Example 4: Reliable vs. Unreliable Source Summary

FactReasoner — Ideation

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1 Synthesizing a summary from a reliable and an unreliable source

This example starts from two source paragraphs about the same subject. Source **A** is largely accurate; source **B** is a sensational news piece containing several factual errors that *contradict* A. We then synthesize a summary **S** that deliberately interleaves claims from both, so that S contains both logical-implication chains and positional invalidations, and we analyse S with the usual decomposition.

Source A (reliable)

Keyara Jones is an American college basketball player who played in the NCAA Division I women’s basketball league. Jones played college basketball for the University of Alabama during the 2020–2021 season, as a member of the Crimson Tide women’s basketball team, which competes in the Southeastern Conference (SEC) of NCAA Division I. The NCAA Division I women’s basketball league consists of 356 teams and is considered the highest level of women’s college basketball in the United States. According to the University of Alabama’s athletic website, Jones is listed as a *guard from Birmingham, Alabama*, and was a member of the women’s basketball team during the 2020–2021 season; ESPN and CBS Sports also reported on her performances that season.

Source B (unreliable — contains factual errors)

“Breaking News: Alabama’s Keyara Jones Makes History as First Female Player in NCAA Division I *Men’s* Basketball.” The 20-year-old junior, a 5’10” *guard from Montgomery, Alabama*, made her debut last night. Averaging 18 points and 7 rebounds per game, she led the Alabama women’s team to a Sweet 16 appearance last season. Coach Nate Oats praised her, and a sports sociologist called it a milestone for gender equality. “According to NCAA data, only 12% of Division I men’s basketball players are women,” the piece reports, framing Jones’ entry as a step toward greater gender representation; several NBA teams are said to be scouting her.

Errors in B (each contradicts A). (i) Jones joined an NCAA Division I *men’s* team — A has her on the *women’s* team. (ii) She is a guard from *Montgomery* — A’s cited source says *Birmingham*. (iii) The “12% of men’s players are women” statistic is internally absurd. The fabricated junior/height/Sweet 16/NBA-scouting details have no support in A.

Synthesized summary S

S is written to interleave B’s sensational (false) framing *first* with A’s reliable record *afterward*, so that the later (true) atoms positionally invalidate the earlier (false) ones, while several implication chains run within each block.

Plain text.

Keyara Jones is an American college basketball player from Montgomery, Alabama, who plays guard for the University of Alabama. During the 2020–2021 season she made history as the first woman to join an NCAA Division I men’s basketball team, averaging 18 points per game and leading Alabama to a Sweet 16 appearance. Because she competed in the men’s league, her achievement was hailed as a milestone for gender equality, and NCAA data showing that 12% of Division I men’s players are women was cited to underscore the trend. In fact, the University of Alabama’s athletic website lists Jones as a guard from Birmingham, Alabama, who was a member of the women’s basketball team during the 2020–2021 season. The Crimson Tide women’s team competes in the Southeastern Conference of NCAA Division I women’s basketball, which is the highest level of women’s college basketball in the United States and consists of 356 teams.

Annotated. Each clause is one atomic unit S^n . Red marks a false atom imported from source B, green a reliable atom from source A; neutral (blue) atoms are uncontested.

Keyara Jones is an American college basketball player^{S1} from Montgomery, Alabama^{S2}, who plays guard for the University of Alabama^{S3}. During the 2020–2021 season she made history as the first woman to join an NCAA Division I men’s basketball team^{S4}, averaging 18 points per game^{S5} and leading Alabama to a Sweet 16 appearance^{S6}. Because she competed in the men’s league^{S7}, her achievement was hailed as a milestone for gender equality^{S8}, and NCAA data showing that 12% of Division I men’s players are women^{S9} was cited to underscore the trend^{S10}. In fact, the University of Alabama’s athletic website lists Jones as a guard from Birmingham, Alabama^{S11}, who was a member of the women’s basketball team during the 2020–2021 season^{S12}. The Crimson Tide women’s team competes in the Southeastern Conference of NCAA Division I women’s basketball^{S13}, which is the highest level of women’s college basketball in the United States^{S14} and consists of 356 teams^{S15}.

Atomic units of S (self-contained, in order of appearance)

- S1.** Keyara Jones is an American college basketball player.
- S2.** Keyara Jones is from Montgomery, Alabama.
- S3.** Keyara Jones plays guard for the University of Alabama.
- S4.** During the 2020–2021 season Keyara Jones made history as the first woman to join an NCAA Division I men’s basketball team.
- S5.** Keyara Jones averaged 18 points per game.
- S6.** Keyara Jones led Alabama to a Sweet 16 appearance.
- S7.** Keyara Jones competed in the men’s basketball league.
- S8.** Keyara Jones’ achievement was hailed as a milestone for gender equality.
- S9.** NCAA data show that 12% of Division I men’s basketball players are women.
- S10.** The 12% statistic was cited to underscore the gender-equality trend.
- S11.** The University of Alabama’s athletic website lists Keyara Jones as a guard from Birmingham, Alabama.

- S12.** Keyara Jones was a member of the University of Alabama women’s basketball team during the 2020–2021 season.
- S13.** The Crimson Tide women’s team competes in the Southeastern Conference of NCAA Division I women’s basketball.
- S14.** NCAA Division I women’s basketball is the highest level of women’s college basketball in the United States.
- S15.** NCAA Division I women’s basketball consists of 356 teams.

Relations

Logical prerequisites ($A \rightarrow B$).

- $S3 \rightarrow S4$ — playing for Alabama is presupposed by joining an Alabama (men’s) team.
- $S4 \rightarrow S7$ — joining a men’s team is what it means to compete in the men’s league.
- $S7 \rightarrow S8$ — competing in the men’s league is the stated ground for the gender-equality acclaim.
- $S9 \rightarrow S10$ — the 12% statistic is what is cited to underscore the trend.
- $S8, S10$ share the gender-equality framing ($S10$ reinforces $S8$).
- $S12 \rightarrow S13$ — membership of the women’s team is presupposed by that team competing in the SEC.
- $S13 \rightarrow S14$ — the women’s league existing/being top-level is presupposed by $S13$ placing the team in it.
- $S14 \rightarrow S15$ — the league being characterised (top level) is presupposed by counting its 356 teams.
- $S1 \rightarrow S3$ — being a college basketball player is presupposed by playing guard for a specific program.

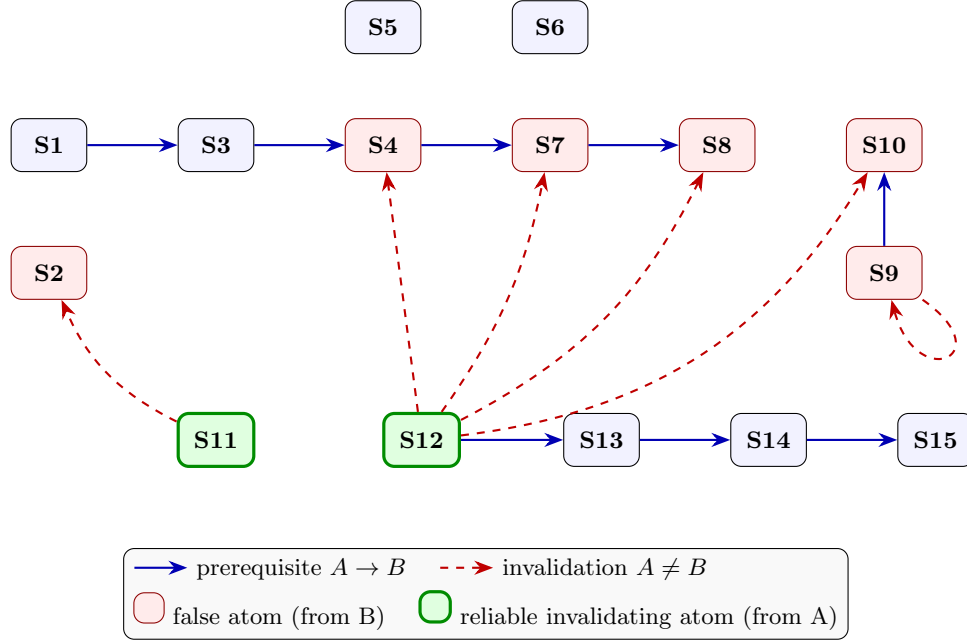
Position invalidations ($A \neq B$). The reliable atoms $S11$ and $S12$ (from source A), appearing *after* B’s framing, positionally invalidate the earlier false atoms.

- $S11 \neq S2$ — the cited website says *Birmingham*, contradicting “from Montgomery.”
- $S12 \neq S4$ — being on the *women’s* team contradicts “first woman on a men’s team.”
- $S12 \neq S7$ — women’s-team membership contradicts “competed in the men’s league.”
- $S12 \neq S8$ — with no men’s-league entry, the gender-equality milestone collapses.
- $S12 \neq S10$ — the men’s-representation framing is voided once the men’s-league premise fails.
- $S9 \neq S9$ (self-inconsistent) — “12% of *men’s* players are women” is internally contradictory regardless of position.

Note. The Sweet 16 and 18-ppg atoms ($S5, S6$) are unsupported by A but not directly contradicted by it; they are left as neither implied nor invalidated. The contradiction structure is concentrated on $S12$ (true women’s-team membership) overturning the men’s-league thread $S4$ – $S10$.

Relation graph

Solid blue arrows are logical prerequisites ($A \rightarrow B$); dashed red arrows are position invalidations ($A \neq B$). False atoms imported from B are tinted red; the two reliable invalidating atoms from A (S11, S12) are green and highlighted. The self-contradictory statistic S9 carries a red self-loop.



(S5 and S6 — the unsupported 18-ppg and Sweet 16 claims — are shown without relation edges: A neither implies nor contradicts them.)